

Sum of independent Gamma distributions is a Gamma distribution

Asked 11 years, 11 months ago Modified 4 years, 1 month ago Viewed 125k times



If $X \sim \Gamma(a_1, b)$ and $Y \sim \Gamma(a_2, b)$, I need to prove $X + Y \sim \Gamma(a_1 + a_2, b)$ if X and Y are independent.

38

I am trying to apply formula for independence integral and just trying to multiply the gamma function but stuck ?



probability probability-distributions gamma-distribution

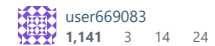


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edited Sep 18, 2020 at 16:07



asked Dec 3, 2012 at 16:16



Hint: After multiplying $f_{X_1}(x)$ and $f_{X_2}(z-y)$ and making sure that the limits are correct, you will get an integral for $f_{X_1+Y_2}(z)$ that can be transformed into a *Beta* function whose value is $B(a_1, a_2) = \frac{\Gamma(a_1)\Gamma(a_2)}{\Gamma(a_1+a_2)}$. – Dilip Sarwate Dec 3, 2012 at 16:28

3 Answers

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Now that the homework deadline is presumably long past, here is a proof for the case of $b = 1$, adapted from an [answer](#) of mine on stats.SE, which fleshes out the details of what I said in a comment on the question.

41

If X and Y are independent continuous random variables, then the probability density function of $Z = X + Y$ is given by the convolution of the probability density functions $f_X(x)$ and $f_Y(y)$ of X and Y respectively. Thus,

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx.$$

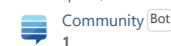


But when X and Y are nonnegative random variables, $f_X(x) = 0$ when $x < 0$, and for positive number z , $f_Y(z-x) = 0$ when $x > z$. Consequently, for $z > 0$, the above integral can be simplified to

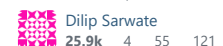
$$\begin{aligned} f_{X+Y}(z) &= \int_0^z f_X(x) f_Y(z-x) dx \\ &= \int_0^z \frac{x^{a_1-1} e^{-x}}{\Gamma(a_1)} \frac{(z-x)^{a_2-1} e^{-(z-x)}}{\Gamma(a_2)} dx \\ &= e^{-z} \int_0^z \frac{x^{a_1-1} (z-x)^{a_2-1}}{\Gamma(a_1)\Gamma(a_2)} dx && \text{now substitute } x=zt \text{ and think} \\ &= e^{-z} z^{a_1+a_2-1} \int_0^1 \frac{t^{a_1-1} (1-t)^{a_2-1}}{\Gamma(a_1)\Gamma(a_2)} dt && \text{of Beta}(a_1, a_2) \text{ random variables} \\ &= \frac{e^{-z} z^{a_1+a_2-1}}{\Gamma(a_1 + a_2)} \end{aligned}$$

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edited Apr 13, 2017 at 12:44



answered Mar 14, 2013 at 18:23



2 Dilip, Your response assumes that parameter $B = 1$ is the same for both distributions. What would the function for the sum of two independent Gamma distributions be where both sets of parameters A and B are different? Convolute the two functions $z^{A_i-1} \text{Exp}(-B_i z) / \text{Gamma}(A_i)$, where i is index 1 and 2. – Tomas Kollen Mar 14, 2014 at 21:26

1 I wonder what the down voter found so objectionable about this answer. – Dilip Sarwate Nov 7, 2017 at 3:17

3 You waited till his homework deadline is presumably long past to answer?! How cruel. – A_for_Abacus Mar 11, 2018 at 20:33

8 @A_for_Abacus I gave a *hint* (actually a complete sketch of the answer) about the homework problem 12 minutes after the question was posted by the OP. So, there was no cruelty involved. – Dilip Sarwate Mar 12, 2018 at 0:19

Hey guys, I'm confusing why the integral started from 0 to z not 0 to ∞ ? – ToadetteK May 11, 2023 at 8:50



It's easier to use Moment Generating Functions to prove that.

22

$$\begin{aligned} M(t; \alpha, \beta) &= Ee^{tX} = \int_0^{+\infty} e^{tx} f(x; \alpha, \beta) dx = \int_0^{+\infty} e^{tx} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} dx \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^{+\infty} x^{\alpha-1} e^{-(\beta-t)x} dx = \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{\Gamma(\alpha)}{(\beta-t)^\alpha} = \frac{1}{(1 - \frac{t}{\beta})^\alpha} \end{aligned}$$



By using the property of independent random variables, we know

$$M_{X+Y}(t) = M_X(t) M_Y(t)$$

So if $X \sim \text{Gamma}(\alpha_1, \beta)$, $Y \sim \text{Gamma}(\alpha_2, \beta)$,

$$M_{X+Y}(t) = \frac{1}{(1 - \frac{t}{\beta})^{\alpha_1}} \frac{1}{(1 - \frac{t}{\beta})^{\alpha_2}} = \frac{1}{(1 - \frac{t}{\beta})^{\alpha_1 + \alpha_2}}$$

You can see the MGF of the product is still in the format of Gamma distribution. Finally we can get $X + Y \sim \text{Gamma}(\alpha_1 + \alpha_2, \beta)$

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edited Oct 2, 2020 at 22:48
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1

answered Nov 6, 2016 at 20:18
 Albert Chen
497 3 9

▲
21

You may use a easier method. Consider the moment generating function or probability generating function. $E(e^{(X+Y)t}) = E(e^{Xt}e^{Yt}) = E(e^{Xt})E(e^{Yt})$ as they are independent then we can get a moment generating function of a gamma distribution. Then you can find the mean and variance from the Moment generating function

▼



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edited Apr 21, 2014 at 2:04
 user2345215
16.6k 2 31 58

answered Dec 3, 2012 at 16:28
 Mathematics
4,591 6 35 62